MODULE 08: INDUCTION & INDUCTANCE

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LECTURE 17

OUTLINE:
- Faraday's law of induction
- Lenz's law
- Induction and energy transfer

BACKGROUND

- For a long time, ***Electricity and Magnetism were considered separate and unrelated phenomena***.
- In early decades of nineteenth century, experiments on electric current by ***Oersted*** demonstrated that, when the current flows through a conducting wire, it produces magnetic field around it.
  - Compass needle moves in response to magnetic field when placed near a current carrying wire.
  - This established the fact that electricity and magnetism are interrelated.
- Later on, Faraday showed that the ***change in magnetic field can create current in a coil***, known as electromagnetic induction.

ELECTROMAGNETIC INDUCTION

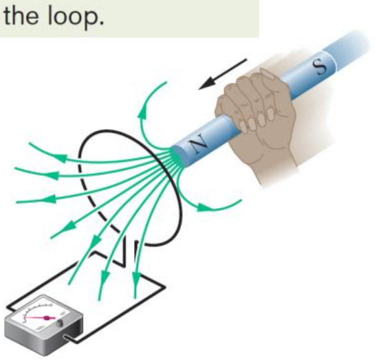
Faraday and Henry conducted several experiments to understand the electromagnetic induction.

First Experiments:

Let us consider a "Conducting wire loop" that is connected to the galvanometer/ammeter, and this conducting loop is approached with a bar magnet. The generation of current through the loop is observed, due to the back-and-forth movement of the magnet perpendicular to the loop area.

- When a magnet moves towards the loop, the defection of the ammeter is observed.
  - Due to the flow of current through the loop.
- When the magnet is held stationary, there is no defection in the ammeter.
  - Thus, no induce current produced.
  - Even though the magnet is in the loop.
- Finally, when the magnet moves away from the loop,
  - The ammeter deflects again, but in the opposite direction.
  - Which indicates the current generation in opposite direction.

The magnet's motion creates a current in the loop.

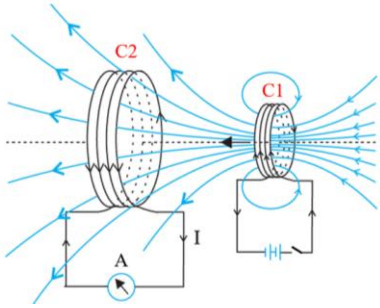


- The current that generates due to change in magnetic field through the loop is known as "***Induced Current***'.

Second Experiment:

Faraday performed lots of experiments. In one of the experiments,

- A magnetic field is produced by running a current through a loop (C_1) using battery, which creates a magnetic field.
  - By switching the current on and off, leading to the change in magnetic field generation.
- A second conducting wire loop (C_2) that is connected to the ammeter (Fig.), detects the current flow through the coil C_2.
- Experiment shows that the EMF that is generated in the second coil (C_2) is proportional to the change of magnetic field (dB_1/dt) produced by coil C_1.
- In addition, the EMF that is generated in second coil (C_2) is proportional to the area (A_2) of this coil C_2.
- This generates the idea that the EMF is actually related to the change of ***magnetic flux (Φ)*** through the surface of the coil.



Therefore, we get,

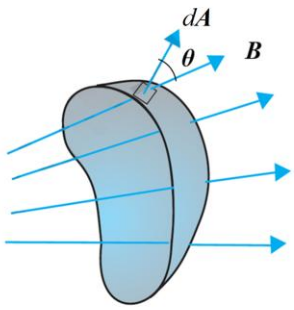
$$\varepsilon_{ind\_2} \propto \frac{dB}{dt}$$

$$\varepsilon_{ind\_2} \propto area\ (A)$$

Magnetic flux: Magnetic flux describes the total magnetic field that passes through a given area.

For an open surface, if the local area vector perpendicular to the surface is $d\vec{A}$, and the local magnetic field is $\vec{B}$, then a magnetic flux (Φ_B) through surface is defined mathematically as,

$$\Phi_B = \iint_{Open\ surface\ (A)} \vec{B}.d\vec{A}$$



Where, $d\vec{A}$ is an area vector having magnitude of dA that is perpendicular to a differential area dA. The $\vec{B}.d\vec{A} = B\ dA\cos\theta$ gives the piercing component, where θ is the angle between $\vec{B}$ and $d\vec{A}$.

- SI unit of magnetic flux is $weber\ (Wb)$. $1\ Wb = 1\ T.m^2$

FARADAY'S LAW

Faraday's law state that the magnitude of the ***induced emf*** in a conducting loop is equal to the time rate of change of magnetic flux through the loop.

$$\varepsilon = -\frac{d\Phi_B}{dt}$$

The negative sign indicates the direction of the induced emf (ε) and hence the direction of the induced current in a closed loop.

- If the magnetic flux is changed through a coil, having N number of turns, an induced emf appears in every turn. And the total emf induced in the coil is the sum of these individual induced emfs. If the coil is tightly wound, so that the same magnetic flux Φ_B passes through all the turns, the total emf induced in the coil is given by,

$$\varepsilon = -N\frac{d\Phi_B}{dt}$$

The induced emf can be increased by increasing the number of turns N of a closed coil.

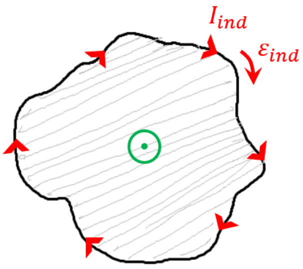
<u>*Induced EMF*</u>: Induced electromotive force (emf), or electromotive force, refers to the voltage or potential difference that is generated in a conductor or closed loop of wire or circuit when there is a change in the magnetic field passing through the conductor.

<u>*Induced current*</u>: Induced current refers to the electric current that is generated in a conductor or closed loop of wire or circuit due to a change in magnetic field passing through the conductor.

- The change in the magnetic field passing through a loop of wire induces an electromotive force (EMF) in the conductor that subsequently drives an electric current.

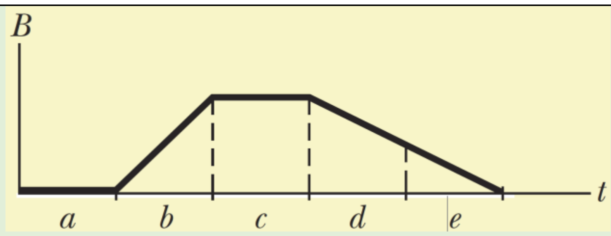
<u>*Direction of the induced EMF*</u>: The direction of an induced electromotive force (emf) or current in a closed electrical circuit depends on the ***increasing*** or ***decreasing magnetic field*** through it. Let us consider, a conducting wire loop, and a magnetic field is coming out of the surface of the loop, and if this field is growing / increasing.

- According to Lenz's law, there is an induced current flowing through loop in a clockwise direction (indicated by red arrows).
- Thus, the EMF (ε) must also be in the clockwise direction.
- In this case, EMF can be defined as closed loop integral of $\vec{E}.d\vec{l}$ *[where, $d\vec{l}$ refers to small distance along with the charge particle move in a straight line]*. Therefore,



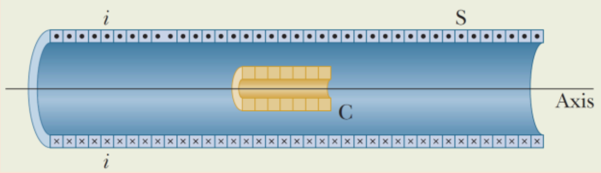
$$\varepsilon_{ind} = \oint_L \vec{E}.d\vec{l}$$

CHECKPOINT: The graph gives the magnitude $B(t)$ of a uniform magnetic field that exists throughout a conducting loop, with the direction of the field perpendicular to the plane of the loop. Rank the five regions of the graph according to the magnitude of the emf induced in the loop, greatest first.



ANSWER: *b*, then *d* and *e* tie, and then *a* and *c* tie (zero)

PROBLEM: The long solenoid S shown (in cross section) in Figure has $220\ turns/cm$ and carries a current $i = 1.5\ A$; its diameter D is $3.2\ cm$. At its center we place a $130-turn$ closely packed coil C of diameter $d = 2.1\ cm$.



The current in the solenoid is reduced to zero at a steady rate in $25\ ms$. What is the magnitude of the emf that is induced in coil C while the current in the solenoid is changing?

ANSWER: We know that, $\varepsilon = -N\frac{d\Phi_B}{dt}$

where the number of turns N is 130 and $\frac{d\Phi_B}{dt}$ is the rate at which the flux changes.

Since, the current in the solenoid decreases at a steady rate, flux Φ_B also decreases at a steady rate, and so we can write $\frac{d\Phi_B}{dt}$ as $\frac{\Delta\Phi_B}{\Delta t}$.

The final flux $\Phi_{B,f}$ is zero because the final current in the solenoid is zero. Here, area $A = \frac{1}{4}\pi d^2$ ($= 8.042 \times 10^{-4}\ m^2$) and the number $n = 220 \frac{turns}{cm} = 22\ 000\ turns/m$. Therefore, we get,

The initial flux $\Phi_{B,i} = BA = (\mu_0 in)A$

$$\Phi_{B,i} = \left(4\pi \times 10^{-7}\ T.\frac{m}{A}\right)(1.5\ A)\left(22000\frac{turns}{m}\right)(8.042 \times 10^{-4}\ m^2) = 3.334 \times 10^{-5}\ Wb$$

Now we can write,

$$\frac{d\Phi_B}{dt} = \frac{\Delta\Phi_B}{\Delta t} = \frac{\Phi_{B,f} - \Phi_{B,i}}{\Delta t} = \frac{(0 - 3.335 \times 10^{-5}\ Wb)}{(25 \times 10^{-3}\ s)}$$

$$\frac{d\Phi_B}{dt} = -1.334 \times 10^{-3}\frac{Wb}{s} = -1.335 \times 10^{-5} V$$

We are interested only in magnitudes; So, we ignore the minus signs here writing

$$\varepsilon = -N\frac{d\Phi_B}{dt} = (130\ turns)(1.335 \times 10^{-5}\ V) = 17.34 \times 10^{-2}\ V$$

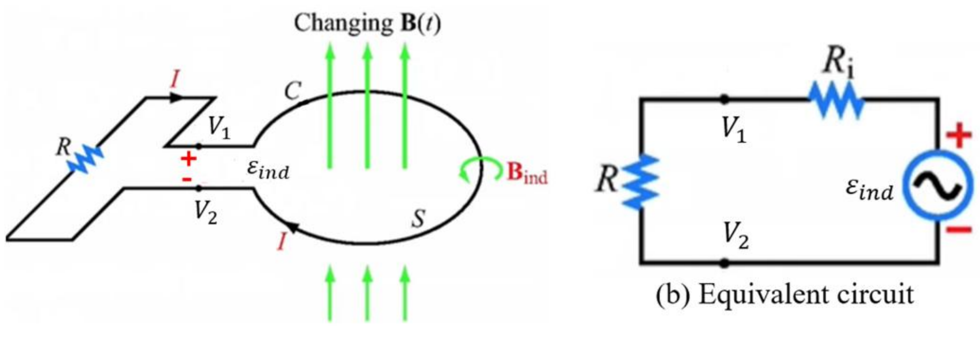
$$\varepsilon = 173.41\ mV$$

LENZ'S LAW: LAW OF CONSERVATION OF ENERGY

Lenz's law states that the induced current that is generated in a conducting loop of wire, due to the change in the magnetic field through a closed loop (or coil), is always in a direction that opposes the change of magnetic flux that produced it. Induced emf (ε_{ind}) can be written as,

$$\varepsilon_{ind} = I_{ind}R \qquad [\because V = iR]$$

Here, I_{ind} is the induced current and R is the resistance of the conducting loop.



(a) Loop in changing **B** field

(b) Equivalent circuit

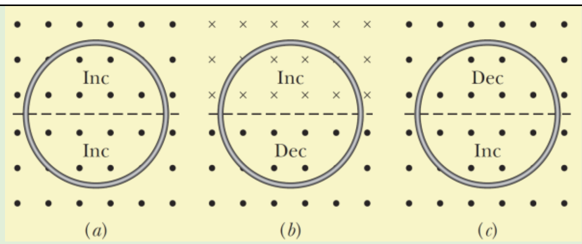
Let us consider a stationary loop in a time varying $\vec{B}$ field, where magnetic fields are perpendicular to the surface attached to the conductive loop.

- This creates an induced current (i) through the loop, and thus, an induced EMF / potential difference between two-side of the loop.
- The polarity of induced emf (ε_{ind}) is such that it tends to produce a current which opposes the change in magnetic flux that produced it.

- In other words, the flow of ***induced current through*** the loop ***produces a secondary magnetic field***, that will oppose the change in primary external magnetic field.
  - This law follows the law of conservation of energy.
  - The induce current can be calculated by dividing induced EMF by the resistance. Therefore, the induced current, $I_{ind} = \frac{\varepsilon_{ind}}{R}$.

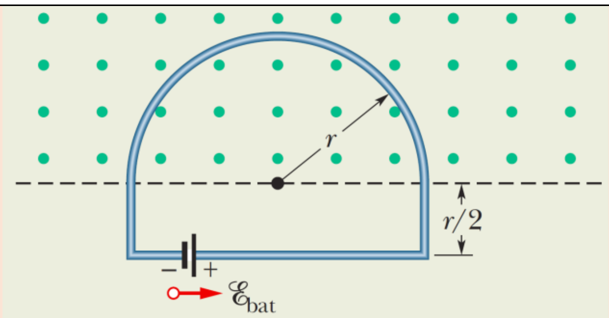
NOTE: Otherwise, if this is not the case, generation of induced current will increase the magnetic flux, since the secondary magnetic flux would be additive to the primary flux. The increase of magnetic flux will then increase the induce which will then increase the flux more… and so on.

CHECKPOINT: The figure shows three situations in which identical circular conducting loops are in uniform magnetic fields that are either increasing (Inc) or decreasing (Dec) in magnitude at identical rates. In each, the dashed line coincides with a diameter. Rank the situations according to the magnitude of the current induced in the loops, greatest first.



ANSWER: a and b tie, then c (zero)

PROBLEM: Figure shows a conducting loop consisting of a half-circle of radius $r = 0.20\ m$ and three straight sections. The halfcircle lies in a uniform magnetic field $\vec{B}$ that is directed out of the page; the field magnitude is given by $B = 4.0t^2 + 2.0t + 3.0$, with B in teslas and t in seconds. An ideal battery with emf $\varepsilon_{bat} = 2.0\ V$ is connected to the loop. The resistance of the loop is 2.0 Ω.



(a) What are the magnitude and direction of the emf ε_{ind} induced around the loop by field at $t = 10\ s$? (b) What is the current in the loop at $t = 10\ s$?

ANSWER: (a) The field magnitude B changes in time (not the area A) we get,

$$\varepsilon_{ind} = \frac{d\Phi_B}{dt} = \frac{d(BA)}{dt} = A\frac{dB}{dt}$$

Because the flux penetrates the loop only within the half-circle, the area A in this equation is $\frac{1}{2}\pi r^2$. Substituting this and the given expression for B yields,

$$\varepsilon_{ind} = A\frac{dB}{dt} = \frac{\pi r^2}{2}\frac{d}{dt}(4.0t^2 + 2.0t + 3.0) = \frac{\pi r^2}{2}(8.0t + 2.0)$$

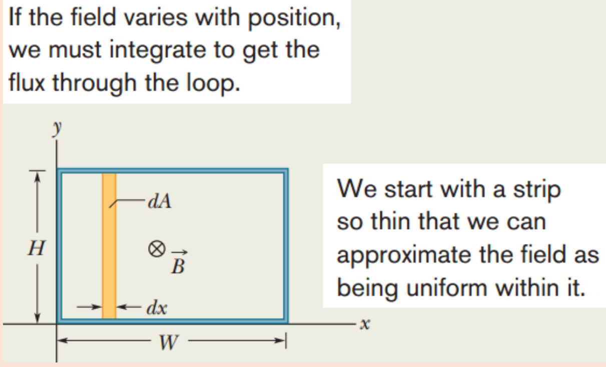
At $t = 10\ s$, then

$$\varepsilon_{ind} = \frac{\pi(0.20\ m)^2}{2}[(8.0)(10\ s) + 2.0] = 5.152\ V \approx 5.2\ V$$

(b) The induced emf ε_{ind} tends to drive a current clockwise around the loop; the battery's emf ε_{bat} tends to drive a current counterclockwise. Because ε_{ind} is greater than ε_{bar}, the net emf ε_{net} is clockwise, and thus so is the current. To find the current at $t = 10\ s$, we use equation ($i = \varepsilon/R$):

$$i = \frac{\varepsilon_{net}}{R} = \frac{\varepsilon_{ind} - \varepsilon_{bat}}{R} = \frac{5.152\ V - 2.0\ V}{2.0\ \Omega} = 1.58\ A \approx 1.60\ A$$

PROBLEM: Figure shows a rectangular loop of wire immersed in a nonuniform and varying magnetic field $\vec{B}$ that is perpendicular to and directed into the page. The field's magnitude is given by $B = 4t^2x^2$, with B in teslas, t in seconds, and x in meters. The loop has width $W = 3.0\ m$ and height $H = 2.0\ m$. What are the magnitude and direction of the induced emf # around the loop at $t = 0.10\ s$?



ANSWER: In Fig., $\vec{B}$ is perpendicular to the plane of the loop (thus, parallel to the differential area vector $d\vec{A}$); So, the dot product gives $B\ dA$. As, the magnetic field varies with the coordinate x but not with the coordinate y, we can take the differential area dA to be the area of a vertical strip of height H and width dx (as shown in Fig). Then $dA = H\ dx$, and the flux through the loop is,

$$\Phi_B = \int \vec{B}.d\vec{A} = \int B\ dA = \int BH\ dx = \int 4t^2x^2\ H\ dx$$

Here, $t = constant$, and inserting the integration limits $x = 0$ and $x = 3.0\ m$, we obtain

$$\Phi_B = 4t^2H \int_0^{3.0} x^2\ dx = 4t^2H \left[\frac{x^3}{3}\right]_0^{3.0} = 72t^2$$

where we have substituted $H = 2.0\ m$ and Φ_B is in webers. Now we can use Faraday's law to find the magnitude of ε at any time *t*,

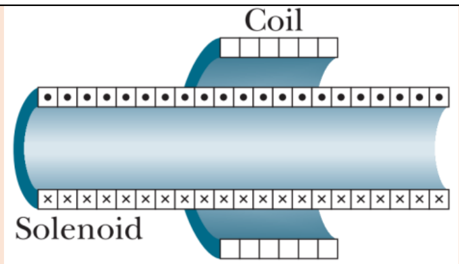
$$\varepsilon = \frac{d\Phi_B}{dt} = \frac{d(72t^2)}{dt} = 144\ t$$

in which ε is in volts. At $t = 0.10\ s$,

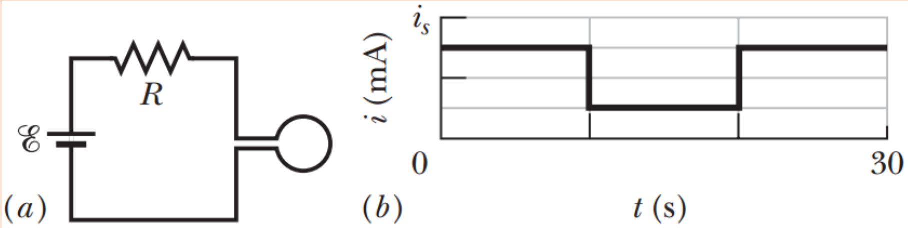
$$\varepsilon = (144\frac{V}{s})(0.10\ s) \approx 14\ V$$

PROBLEM 30-02: A certain elastic conducting material is stretched into a circular loop of $12.0\ cm$ radius. It is placed with its plane perpendicular to a uniform $0.800\ T$ magnetic field. When released, the radius of the loop starts to shrink at an instantaneous rate of $75.0\ cm/s$. What emf is induced in the loop at that instant?

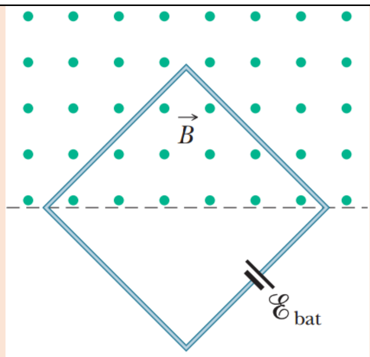
PROBLEM 30-03: In Fig. a $120 - turn$ coil of radius $1.8\ cm$ and resistance $5.3\ \Omega$ is coaxial with a solenoid of $220\ turns/cm$ and diameter $3.2\ cm$. The solenoid current drops from $1.5\ A$ to zero in time interval $\Delta t = 25\ ms$. What current is induced in the coil during Δt?



PROBLEM 30-06: Figure a shows a circuit consisting of an ideal battery with emf $\varepsilon = 6.00\ \mu V$, a resistance R, and a small wire loop of area $5.0\ cm^2$. For the time interval $t = 10\ s$ to $t = 20\ s$, an external magnetic field is set up throughout the loop. The field is uniform, its direction is into the page in Fig. a, and the field magnitude is given by $B = at$, where B is in teslas, a is a constant, and t is in seconds. Figure b gives the current i in the circuit before, during, and after the external field is set up. The vertical axis scale is set by $i_s = 2.0\ mA$. Find the constant a in the equation for the field magnitude.

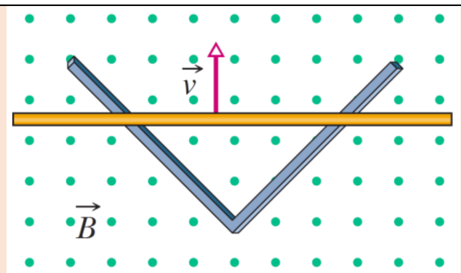


PROBLEM 30-15: A square wire loop with $2.00\ m$ sides is perpendicular to a uniform magnetic field, with half the area of the loop in the field as shown in Fig. The loop contains an ideal battery with emf $\varepsilon = 20.0\ V$. If the magnitude of the field varies with time according to $B = 0.0420 - 0.870t$, with B in teslas and t in seconds, what are (a) the net emf in the circuit and (b) the direction of the (net) current around the loop?



PROBLEM 30-17: A small circular loop of area $2.00\ cm^2$ is placed in the plane of, and concentric with, a large circular loop of radius $1.00\ m$. The current in the large loop is changed at a constant rate from $200\ A$ to $-200\ A$ (a change in direction) in a time of $1.00\ s$, starting at $t = 0$.What is the magnitude of the magnetic field $\vec{B}$ at the center of the small loop due to the current in the large loop at (a) $t = 0$, (b) $t = 0.500\ s$, and (c) $t = 1.00\ s$? (d) From $t = 0$ to $t = 1.00\ s$, is $\vec{B}$ reversed? Because the inner loop is small, assume $\vec{B}$ is uniform over its area. (e) What emf is induced in the small loop at $t = 0.500\ s$?

PROBLEM 30-18: In Figure, two straight conducting rails form a right angle. A conducting bar in contact with the rails starts at the vertex at time $t = 0$ and moves with a constant velocity of $5.20\ m/s$ along them. A magnetic field with $B = 0.350\ T$ is directed out of the page. Calculate (a) the flux through the triangle formed by the rails and bar at $t = 3.00\ s$ and (b) the emf around the triangle at that time. (c) If the emf is $\varepsilon = at^n$, where a and n are constants, what is the value of n?



PROBLEM 30-20: At a certain place, Earth's magnetic field has magnitude $B = 0.590\ gauss$ and is inclined downward at an angle of 70.0° to the horizontal. A flat horizontal circular coil of wire with a radius of $10.0\ cm$ has $1000\ turns$ and a total resistance of 85.0 Ω. It is connected in series to a meter with 140 Ω resistance. The coil is flipped through a half-revolution about a diameter, so that it is again horizontal. How much charge flows through the meter during the flip?

INDUCTION AND ENERGY TRANSFERS

Induction and energy transfer are closely related concepts in the context of electromagnetism.

Induction: Electromagnetic induction refers to the process of generating an induced electromotive force (EMF) in a closed electrical circuit due to a change in magnetic field passing through the circuit.

- Energy transfer involves the transfer of energy from one system to another.
- The induced EMF can be used to transfer energy from the magnetic field to the conductor.
- The induced current, driven by the induced EMF, allows the transfer of electrical energy within the conductor or to connected devices.

Form of Energy Transfer:

Electric Power Generation: Induction is widely used in electric power generation.

- In a generator, mechanical energy is used to rotate a coil of wire within a magnetic field, inducing an EMF and generating an alternating current (AC).
- The changing magnetic field induces the current, and the current carries electrical energy that can be harnessed and used for various purposes.

Transformers: Transformers use induction to transfer electrical energy between two or more circuits.

- A changing current in one coil, known as the primary coil, creates a changing magnetic field.
- This changing magnetic field induces an EMF in another coil, known as the secondary coil, which is connected to a different circuit.
- Through electromagnetic induction, energy is transferred from the primary coil to the secondary coil with minimal losses.

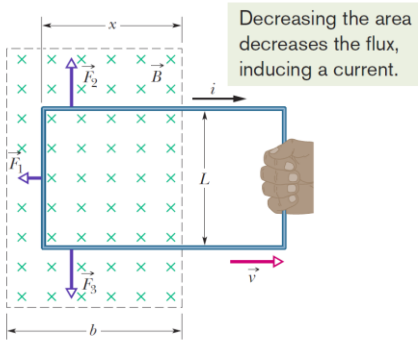
Induction Heating: Induction heating uses induction to transfer heat energy to conductive materials.

- When an alternating current (AC) is passed through an induction coil, it generates a rapidly alternating magnetic field around it.
- If a conductive material, typically a ***metal plate***, is placed within the alternating magnetic field, it experiences a changing magnetic flux.
- This change in magnetic flux induces eddy currents (circulating currents) within the material.
- The induced eddy currents flow through the material and encounter resistance, which results in the material heating up due to Joule heating ($P = i^2R$ heating), where i is the current and R is the resistance of the material.
- This heating process occurs throughout the interior of the material.
- Induction heating can be highly localized because it heats only the region within the proximity of the coil.
- This allows precise and controlled heating without affecting the surrounding material.

Energy Transfer:

Let us consider a rectangular loop of wire with width L. One end of the loop is in a uniform external magnetic field that is directed perpendicularly into the plane of the loop. The dashed lines in Figure show boundary limits of the magnetic field. The fringing of the field at its edges is neglected.

- If the loop is moved to the right at a constant velocity $\vec{v}$.
- The change in magnetic flux passing through a closed conducting loop or coil of wire occurs, due to the relative motion between the magnetic field and the conducting loop.
    - This leads to the induction of an electromotive force (EMF) and subsequently an induced current.



A constant force ($\vec{F_a}$) is required to apply to move the loop with a constant velocity (v), as a magnetic force of equal magnitude but opposite direction acts on the loop to oppose its motion. The rate of work, often referred to as power (P) that is a measure of how quickly work is done or the rate at which energy is transferred or converted, is then given by,

$$P = \frac{dW}{dt} = \frac{F_a dx}{dt} = F_a v$$

Here F_a is the magnitude of the applied force. The SI unit of power is the watt ($1W = 1J/s$).

The magnitude of the flux through the loop is,

$$\Phi_B = BA = BLx$$

As x decreases due to motion of the coil to the right, the flux decreases. According to the Faraday's law as the flux decrease, an emf is induced in the loop. We can write as,

$$\varepsilon_{ind} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(BLx) = -BL\frac{dx}{dt} = -BLv$$

Here dx/dt is replaced with v, the speed at which the loop moves, ε is the induced electromotive force, $\frac{d\Phi_B}{dt}$ is the rate of change of magnetic flux, and the negative sign indicates that the induced EMF opposes the change in flux.

The magnitude of the induced current is given by,

$$i = \frac{|\varepsilon_{ind}|}{R} = \frac{BLv}{R}$$

Since three segments of the loop in Figure carry this current through the magnetic field, sideways deflecting forces ($\vec{F}_d$) act on those segments. The deflecting forces acting on the three segments of the loop are marked with $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$. Since, from the symmetry, forces $\vec{F}_2$ and $\vec{F}_3$ are equal in magnitude and cancel each other. This leaves only force $\vec{F}_1$, which is directed opposite to the applied force $\vec{F}_a$ on the loop and thus is the force opposing the motion. So,

$$\vec{F}_a = -\vec{F}_1$$

We know that such a deflecting force is given by,

$$\vec{F}_d = \vec{F}_a = -\vec{F}_1 = i\vec{L} \times \vec{B}$$

$$|F_a| = |F_1| = iLB \sin 90° = iLB = \left(\frac{BLv}{R}\right) LB = \frac{B^2L^2v}{R} \qquad \left[\because i = \frac{BLv}{R}\right]$$

$$\text{Therefore, } P = F_a v = \frac{B^2L^2v^2}{R}$$

Thermal energy: The induced currents flow through the material and encounter resistance, which leads to the Joule heating ($P = i^2R$ heating), where i is the current and R is the resistance of the material. Therefore,

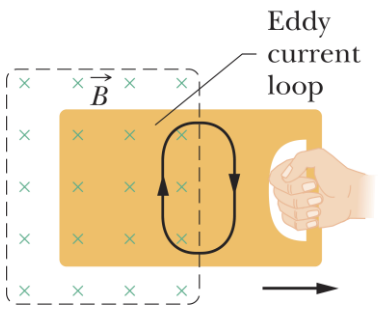
$$P = i^2R = \left(\frac{BLv}{R}\right)^2 R = \frac{B^2L^2v^2}{R}$$

which is exactly equal to the rate at which the work is done on the loop.

Eddy current:

Eddy current is a circulating current that flow in conductive materials (generally bulk) when exposed to a changing magnetic field. These currents are a result of electromagnetic induction.

- Eddy currents flow in closed loops within conductors, in planes perpendicular to the magnetic field.
- Eddy currents create their own magnetic fields, and their direction opposes the change in the original magnetic field that induced them.
- The magnitude and distribution of eddy currents depend on several factors, including the ***material's electrical conductivity***, the ***intensity of the magnetic field***, the ***frequency of the field's change***, and the ***geometry of the conductor***.

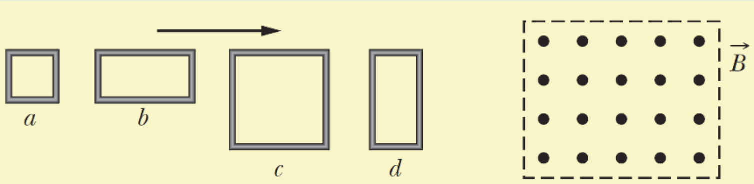


- Eddy currents have both beneficial and undesirable effects. In some applications, such as induction heating and electromagnetic brakes, eddy currents are intentionally generated to produce heating or provide braking action. In these cases, the energy of the eddy currents is transformed into heat.
- However, in many practical scenarios, eddy currents can cause energy losses and undesirable effects. For instance, in transformers and electric motors, eddy currents in the conductive cores can lead to energy dissipation and heat generation, reducing efficiency. To mitigate these effects, laminated or layered cores made of materials with high electrical resistivity are often used to minimize the flow of eddy currents.

Difference between Eddy Current vs. Induced Current:

Eddy current and induced current refer to currents that form on a conductor as a result of a changing magnetic field across it. The main difference between eddy current and induced current is that induced current refers to currents flowing in coils of wire in a closed circuit whereas eddy current refers to loops of currents flowing within pieces of larger conductors due to electromagnetic induction.

CHECKPOINT: The figure shows four wire loops, with edge lengths of either L or $2L$. All four loops will move through a region of uniform magnetic field $\vec{B}$ (directed out of the page) at the same constant velocity. Rank the four loops according to the maximum magnitude of the emf induced as they move through the field, greatest first.



ANSWER: c and d tie, then a and b tie

PROBLEM 30-32: A loop antenna of area $2.00\ cm^2$ and resistance $5.21\ \mu\Omega$ is perpendicular to a uniform magnetic field of magnitude $17.0\ \mu T$. The field magnitude drops to zero in $2.96\ ms$. How much thermal energy is produced in the loop by the change in field?

PROBLEM 30-34: In figure, a long rectangular conducting loop, of width L, resistance R, and mass m, is hung in a horizontal, uniform magnetic field $\vec{B}$ that is directed into the page and that exists only above line aa. The loop is then dropped; during its fall, it accelerates until it reaches a certain terminal speed v_t. Ignoring air drag, find an expression for v_t.